THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 6 — MATHS FULL SYLLABUS PAPER SET 02

Full Syllabus (Chapters 1–10) | RBSE Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. The sum of the first 7 odd numbers is: 1

(a) 42 **(b)** 14 **(c)** 56 **(d)** 49

2. In the Virahānka (Fibonacci) sequence 1, 2, 3, 5, 8, 13, … each term after the first two is: 1

(a) the sum of the two terms before it **(b)** the difference of the previous two
(c) double the previous term **(d)** the product of the previous two

3. Two rays make an angle. The two rays are called its: 1

(a) arms **(b)** edges **(c)** bases **(d)** vertices

4. The angle between North and East is: 1

(a) 180° **(b)** 45° **(c)** 270° **(d)** 90°

5. The Collatz sequence starting at 9 takes how many steps to reach 1? 1

(a) 22 **(b)** 19 **(c)** 18 **(d)** 20

6. The Kaprekar routine on 3524: 5432 − 2345 = 1

(a) 3187 **(b)** 3077 **(c)** 3087 **(d)** 2087

7. In a pictograph where one symbol stands for 3, how many symbols are needed for 18? 1

(a) 6 symbols **(b)** 7 symbols **(c)** 5 symbols **(d)** 8 symbols

8. The data collected from every student of a class asking favourite fruit is an example of: 1

(a) a pictograph key **(b)** collecting data by a survey
(c) drawing a bar **(d)** an angle

9. The process of writing a number as a product of primes is: 1

(a) estimation **(b)** division
(c) prime factorisation **(d)** rounding

10. The HCF of 8 and 20 is: 1

(a) 4 **(b)** 8 **(c)** 3 **(d)** 5

11. The area of two congruent triangles that together form a rectangle is: 1

(a) twice the rectangle **(b)** half the rectangle's area each
(c) one fourth **(d)** equal to the rectangle

12. Area of a rectangle of length 12 cm and breadth 7 cm is: 1

(a) 38 cm² **(b)** 96 cm² **(c)** 19 cm² **(d)** 84 cm²

13. A fraction whose numerator is less than its denominator is a: 1

(a) proper fraction **(b)** improper fraction
(c) mixed fraction **(d)** whole number

14. The improper fraction $\frac{11}{4}$ as a mixed fraction is: 1

(a) $2\frac{3}{4}$ **(b)** $1\frac{3}{4}$ **(c)** $3\frac{1}{4}$ **(d)** $2\frac{1}{4}$

15. A rectangle of sides 15 cm and 8 cm has perimeter: 1

(a) 48 cm **(b)** 46 cm **(c)** 120 cm **(d)** 23 cm

16. The diagonals of a square divide it into: 1

(a) four congruent right triangles **(b)** five parts
(c) three triangles **(d)** two squares

17. The number of lines of symmetry of a regular heptagon is: 1

(a) 1 **(b)** 3 **(c)** 7 **(d)** 14

18. When a figure is reflected in a mirror, its size: 1

(a) changes randomly **(b)** halves **(c)** stays the same **(d)** doubles

19. The value of $-7 + (7)$ is: 1

(a) 2 **(b)** −2 **(c)** 1 **(d)** 0

20. The opposite of +5 is: 1

(a) −5 **(b)** 0 **(c)** +10 **(d)** 5

SECTION — B (Very Short Answer Type) — 8 × 2 = 16

Answer in about 10–15 words each.

21. Add two consecutive triangular numbers 15 and 21. What do you get? 2

22. State the 1089 trick in one line. 2

23. Which is easier to read: a table or a bar graph? Why? 2

24. What are twin primes? Give a pair. 2

25. Find the length of ribbon needed to go round a rectangular card 18 cm by 10 cm. 2

26. Write the like fractions among $\frac{2}{7}, \frac{3}{5}, \frac{5}{7}, \frac{1}{7}$. 2

27. A circle has radius 4 cm. Find its diameter. 2

28. Is zero a positive integer? Give a reason. 2

SECTION — C (Short Answer Type) — 8 × 3 = 24

Answer in about 30–40 words each.

29. Write a rule of your own for a number pattern and give its first five terms. 3

30. Why is the measure of an angle independent of the length of its arms? 3

31. A school survey: 15 prefer tea, 25 prefer milk and 10 prefer juice. Which is the most popular and what fraction prefer milk? 3

32. The perimeter of a triangle is 30 cm and two sides are 9 cm and 11 cm. Find the third side. 3

33. If the diagonals of a quadrilateral bisect each other at 90° and are equal, what is it? 3

34. How can you make a symmetrical design by folding and cutting paper? 3

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35. Which of 221, 223, 225 are prime? Give reasons. 3

OR

What are the different kinds of numbers: prime, composite, co-prime, twin primes? Give an example each. 3

36. Using tokens explain $(-4) - (-6)$. 3

OR

Find the integer which when added to −9 gives 4. 3

SECTION — D (Long Answer Type) 4 × 5 = 20

Answer in about 80–100 words each.

37. **Case study:** A student turns a rotating arm through 45°, 90°, 135° and 180° in turn. 5

(i) Name the types of the angles.
(ii) What fraction of a full turn is 135°?
(iii) How many steps of 45° complete a full turn?
(iv) Which angle is a right angle?

OR

(i) What are collinear points? 5
(ii) How many lines can pass through 2 points?
(iii) How many through 3 non-collinear points taken in pairs?

38. (i) Write numbers between 1,000 and 2,000 whose digits sum to 3. 5
(ii) How many?
(iii) Which is the smallest?

OR

Case study: In a row of numbers 12, 40, 25, 31, 18, a cell is a supercell if it is larger than its adjacent cells. 5

(i) What is a supercell?
(ii) Which numbers are supercells?
(iii) How many are there?
(iv) Will the largest number in any row always be a supercell? Why?

39. (i) Why are $\frac{2}{4}$ and $\frac{1}{2}$ equal? 5
(ii) Show with folding paper.
(iii) Find the fraction equal to $\frac{1}{2}$ with denominator 18.

OR

Case study: A class of 40: $\frac{3}{8}$ walk, $\frac{1}{4}$ cycle, rest use a bus. 5

(i) How many walk?
(ii) How many cycle?
(iii) Fraction using a bus?
(iv) How many use the bus?

40. (i) Complete the figure given its left half and a vertical mirror line. 5
(ii) State the symmetry.
(iii) Name a real object like it.

OR

(i) Explain the symmetry of a regular polygon with n sides. 5
(ii) Apply for n = 8.

END OF QUESTION PAPER Total: 80 Marks